

Customer Churn Prediction

Machine Learning Capstone Project

Predict customers who are likely to leave and support proactive retention decisions.

1. Business Background

Customer retention is an important business priority. Customers may stop using a company's products or services because of changing needs, low engagement, poor experience, competitive alternatives, or other factors. Identifying customers at risk of churn can help an organization take timely retention actions and make better customer-management decisions.

2. Problem Statement

The objective of this project is to develop a Machine Learning classification model that predicts whether a customer is likely to churn based on demographic, financial, and behavioral information. The solution should also help identify the factors associated with churn and provide a reusable pipeline for predicting churn for new customers.

3. Project Objectives

- Explore and understand customer data and churn patterns.
- Perform data preprocessing and feature engineering.
- Train and compare multiple classification algorithms.
- Evaluate models using appropriate classification metrics and cross-validation.
- Select the best-performing model.
- Analyze feature importance where supported by the selected model.
- Save the final preprocessing and prediction pipeline for reuse.

4. Dataset Overview

The dataset contains customer-level information such as **credit score, country, gender, age, tenure, balance, number of products, credit-card status, active-member status, and estimated salary**. The target variable is **churn**, where the model determines whether a customer is predicted to leave or remain.

5. Machine Learning Approach

Data Inspection & EDA → Feature Engineering → Preprocessing → Train/Test Split → Model Training → Cross-Validation → Model Comparison → Final Evaluation → Feature Importance → Pipeline Saving → New Customer Prediction.

6. Expected Outcome

The final solution will generate a churn prediction for a customer and, when supported by the selected classifier, a churn probability. These predictions can support targeted retention strategies, customer prioritization, and data-driven business decisions.